

VCA Video Creation Engine.

1. Automation Project Overview and Objectives

Business Problem Statement

The client currently relies on a completely manual video production process that requires staff to create PowerPoint presentations, manually convert them to video format, add music scores, and produce final training videos for hosting. This manual approach is time-intensive, inconsistent in quality, and creates a significant bottleneck in the client's training content delivery pipeline.

Primary Automation Objectives

- Eliminate manual video production bottlenecks by automating the entire pipeline from script input to final video output
- Achieve 3-4x faster production speed compared to the current manual PowerPoint-based process
- Standardize video quality and branding across all training content through automated template application
- Reduce production timeline from current manual process to 30-minute automated turnaround
- Enable scalable training content creation that can handle multiple simultaneous video production requests

Success Criteria

- Production Time: Complete training video generation within 30 minutes from script input
- Quality Consistency: 100% brand compliance and professional presentation standards across all outputs
- Process Automation: Zero manual intervention required between script input and final video output
- Output Variety: Multi-format generation (MP4, WebM) with multiple resolution options
- Accessibility: Automatic caption and audio description generation for all videos

High-Level Scope Boundaries

The automation encompasses script ingestion, AI avatar generation, voice synthesis, multi-layer composition, and automated output processing. Asset creation (avatars, backgrounds, opener/closer videos) and initial video concept development are handled separately. The system focuses on the production workflow automation that transforms prepared scripts and assets into finished training videos.

2. Target Users and Stakeholders

Primary Users - Client's Content Production Team

- Current Challenge: Manual PowerPoint creation and video conversion processes consuming significant time and resources
- Post-Automation Benefit: Simple script input with automated professional video output, allowing focus on content strategy rather than production mechanics
- Workflow Change: From multi-step manual process to single-input automated pipeline

Secondary Users - Client's Training Department

- Current Challenge: Dependency on production team availability for training video updates and new content creation
- Post-Automation Benefit: Faster turnaround for urgent training content and ability to iterate on video content rapidly
- Workflow Change: Reduced lead times for training material deployment

Stakeholders - Client's Management Team

- Interest: Cost reduction and improved content delivery speed for training programs
- Benefit: Significant resource allocation improvement and enhanced training program scalability
- Success Metric: ROI through reduced production time and improved training content consistency

3. Core Processes, Functionality, and Automation Scope

Current vs. Future Process Flow

Current Manual Process:

1. Create PowerPoint presentation
2. Convert PowerPoint to video format
3. Add music score manually
4. Output and quality check final video
5. Prepare for hosting

Future Automated Process:

1. Input: Training script, audio script, segment content, and illustration ideas
2. Automated: AI avatar generation with voice synthesis
3. Automated: Multi-layer composition with background integration
4. Automated: Music integration and audio mixing
5. Automated: Quality control validation and multi-format output

Three-Phase Production Workflow

Phase 1: Introduction Sequence

- Full-size avatar generation with realistic AI video avatars
- Voice synthesis from script using cloned voice sample
- Background integration with branded graphics
- Opener video merge with avatar introduction
- Dynamic music integration with automated fade transitions

Phase 2: Training Content

- Avatar scaling animation to circular viewport
- Multi-layer composition with background graphics, line art overlays, and synchronized text
- Content synchronization between on-screen elements and avatar narration
- Automated segment transitions between training modules
- Background music mixing at optimized levels for narration clarity

Phase 3: Closing Sequence

- Avatar expansion animation back to full-size presentation
- Closing integration with provided closer video content
- Final branding application
- Music crescendo automation for professional outro

Key Decision Points and Business Rules

- Avatar Sizing Logic: Automatic scaling based on content phase (full-size for intro/outro, circular for training content)
- Audio Mixing Rules: Dynamic volume adjustment (100% music during opener, 15-20% during training, crescendo for outro)
- Content Synchronization: AI-powered timing alignment between text elements and avatar narration
- Quality Validation: Automated checks for lip-sync accuracy, audio balance, and brand compliance

Volume and Timing Expectations

- Production Timeline: 30-minute turnaround from script input to final video output
- Video Length: 10-minute target length with flexible accommodation for varying content lengths
- Concurrent Processing: System designed to handle multiple video production requests simultaneously

4. Integration Architecture and Platform Recommendations

System Integration Requirements

The automation requires integration between AI avatar generation platforms, voice synthesis engines, video composition tools, and quality control systems. The architecture should support seamless data flow from script input through final video output while maintaining professional production standards.

Platform Approach Rationale

- AI Avatar Generation: Integration with enterprise-grade avatar platforms for realistic human-like video generation
- Voice Synthesis: Advanced voice cloning capabilities for consistent narrator voice across all training videos
- Video Composition Engine: Professional-grade multi-layer composition for complex visual element integration
- Quality Control System: Automated validation for technical and brand compliance standards

- Output Processing: Multi-format generation pipeline for various deployment scenarios

Data Flow Architecture

1. Input Layer: Script ingestion with audio script, segment content, and illustration ideas
2. Processing Layer: AI avatar generation, voice synthesis, and multi-layer composition
3. Quality Layer: Automated validation and compliance checking
4. Output Layer: Multi-format video generation with accessibility features

5. Systems and Tools Inventory

Current Client Environment

- Manual Production Tools: PowerPoint, basic video editing software
- Hosting Infrastructure: Video hosting platform for training content delivery
- Content Management: Current script and content development processes

Required Automation Capabilities

- AI Avatar Generation: Realistic video avatar creation with lip-sync capabilities
- Voice Synthesis: High-quality voice cloning and speech generation
- Video Composition: Multi-layer visual element integration and animation
- Audio Processing: Dynamic music mixing and audio balance optimization
- Quality Control: Automated validation for technical and presentation standards
- Format Processing: Multi-resolution and multi-format output generation

Integration Dependencies

- Asset Integration Framework: Compatibility with separately created avatars, backgrounds, and opener/closer videos

- Script Processing: Ability to parse and segment training scripts for automated production
- Brand Compliance: Integration with brand guidelines for consistent visual presentation
- Accessibility Features: Automated caption and audio description generation

6. Development Milestones

Milestone 1: Core Automation Engine

Business Outcome:

Basic automated video production from script input to simple video output Success Criteria:

Single-phase video generation with AI avatar and voice synthesis

Dependencies:

AI avatar and voice synthesis platform integration

Client Benefit:

Proof of concept demonstrating automation feasibility and basic functionality

Milestone 2: Three-Phase Production Workflow

Business Outcome:

Complete automated workflow with introduction, training content, and closing sequences

Success Criteria:

Full avatar scaling animations, multi-layer composition, and seamless phase transitions

Dependencies:

Background graphics integration and music mixing capabilities Client

Benefit:

Professional-quality training video production matching current manual output standards

Milestone 3: Asset Integration Framework

Business Outcome:

Seamless integration with pre-created assets from companion project

Success Criteria:

Automated consumption of avatars, backgrounds, opener/closer videos, and brand elements

Dependencies:

Asset preparation project completion and standardized asset formats

Client Benefit:

Plug-and-play asset utilization enabling immediate production capability

Milestone 4: Quality Control and Validation System

Business Outcome:

Automated quality assurance ensuring professional presentation standards

Success Criteria:

Audio synchronization validation, visual consistency checks, and brand compliance verification

Dependencies:

Quality control criteria definition and validation algorithm development

Client Benefit:

Consistent professional quality without manual review requirements

Milestone 5: Multi-Format Output Processing

Business Outcome:

Complete production pipeline with multiple output formats and accessibility features

Success Criteria:

MP4/WebM generation, multiple resolutions, automated captions, and audio descriptions

Dependencies:

Output processing pipeline integration and accessibility feature implementation

Client Benefit:

Deployment-ready videos suitable for various hosting platforms and compliance requirements

Milestone 6: Performance Optimization and Deployment

Business Outcome:

Production-ready system achieving 30-minute turnaround target

Success Criteria:

Performance optimization, concurrent processing capability, and deployment package completion

Dependencies:

System testing validation and performance tuning completion

Client Benefit:

Fully operational automation replacing manual PowerPoint-based video production process

About Ovidius

Founded in 2021, Ovidius empowers mid-size and large enterprises across the globe to harness the power of AI. We deliver tangible results, from reduced costs and increased efficiency to a stronger competitive edge. Our team of expert AI strategists, developers, UX designers, and QA specialists becomes an extension of your team, working alongside you to achieve your business objectives.

Delivering Tangible Business Impact

1. **Rapid AI implementation:** Delivering working AI solutions in under 30 days, where clients see tangible results quickly
2. **BPO cost reduction:** Bringing outsourced processes back in-house through AI automation at lower cost
3. **Practical AI experimentation:** Proving real-world use cases through corporate experiments that demonstrate business value
4. **AI+human workflow integration:** Creating effective hybrid processes that combine AI capabilities with human expertise
5. **Resource augmentation:** Executing specialized AI projects beyond clients' internal capabilities
6. **Flexible implementation:** Adapting quickly to changing requirements with a nimble, distributed team working across time zones

Play Well With Others

At Ovidius, collaboration isn't just a buzzword—it's in our DNA. Our deep roots in the marketing and software agency world, starting with our own SEO agency, have shaped our understanding of effective partnerships. We thrive in collaborative environments.

Most of the team at Ovidius has extensive experience working within, for, and alongside marketing and software agencies. Needless to say, we understand what teamwork entails and are adept at incorporating your existing vendors and partners into the value we deliver.

We believe in open communication and seamless coordination- ensuring shared success. We are great when leading specific initiatives that incorporate your internal and external resources, and we are equally adept at serving in a support role for your current relationships- always prioritizing the achievement of your overall objectives.